

E-Brave Career Discovery Platform

Master Source of Truth Document

Final Consolidated Implementation Specification

PART 1: Platform Overview

Vision

To architect the future of career guidance by engineering a multidimensional, scientifically robust assessment system that measures intrinsic aptitude, labor market realities, psychological readiness, and future-skill adaptability, transitioning away from rudimentary career quizzes.

Mission

To provide students, parents, and counselors with a transparent, highly predictive, and engaging platform that maps traits to the modern, automated AI-era economy, yielding actionable insights and educational pathways rather than static job titles.

Objectives

- Minimize student survey fatigue through a dynamic, 100-question hybrid assessment.
- Maintain absolute psychometric quality using a mix of normative and ipsative methodologies.
- Generate highly useful counselor interventions using an automated rule-engine.
- Deliver translated, supportive guidance directly to parents to reduce external academic pressure.
- Identify student archetypes and align them with real-world, future-proofed careers.

Target Audience

Grade 8 through College students (aged 13–22) in both Indian and international educational markets, alongside their parents and institutional school counselors.

Platform Philosophy

The E-Brave platform believes that modern career discovery requires cross-referencing subjective aspirations with academic reality. It focuses heavily on "Future Skills" (AI Readiness, Digital Literacy, Adaptability) because technical roles will evolve, but core competencies insulate students against labor market volatility.

PART 2: Complete Assessment Framework

Assessment Methodology

The assessment employs a dual-scoring psychometric integrity approach. It avoids social desirability bias by utilizing Forced Choice (Ipsative) questions for core vocational matching, Likert Scales (Normative) for rapid baseline mapping, and Situational Judgement Tests (SJTs) for applied future-skills reasoning.

Assessment Structure

100 Total Questions separated into 6 interactive "Labs" completed within 18-20 minutes.

Assessment Sections & Traits Measured

- **1. The Interest Matrix (25 Qs):** Core vocational matching mapping intrinsic motivation across 16 modern sectors (Technology, Healthcare, Business, Finance, Law, Education, Design, Arts, Government, Social Impact, Environment, Research, Skilled Trades, Media, Engineering, Entrepreneurship).
- **2. The Personality Lab (25 Qs):** Evaluates environmental fit using the Big Five (OCEAN) behavioral guardrails (Extroversion, Introversion, Conscientiousness, Emotional Stability, Openness, Leadership, Team Orientation, Independence, Discipline, Adaptability).
- **3. The Future Simulator (20 Qs):** Assesses adaptability and AI-era competencies via SJTs (Critical Thinking, Creativity, Communication, Collaboration, Problem Solving, AI Readiness, Digital Literacy, Learning Agility, Innovation, Decision Making).
- **4. The Reality Check (10 Qs):** Grounds aspirations by evaluating subject strengths, weaknesses, academic performance, learning confidence, and study habits.
- **5. The Contextual Compass (10 Qs):** Evaluates OECD-aligned external factors (Parent Influence, Peer Influence, Social Pressure, Independent Thinking, Exposure to Professionals).
- **6. Career Motivation (10 Qs):** Measures psychological readiness (Career Clarity, Career Confusion, Decision Confidence, Motivation, Ownership).

PART 3: Complete Question Bank

The definitive 100-question taxonomy engineered for high engagement and scientific accuracy.

Q#	Section	Question Text	Answer Options	Type	Trait Measured
1	Interest Matrix	If you had a free weekend to complete any project, which would you choose?	A) Code a website B) Paint/design C) Community cleanup D) Read markets E) Build furniture	Ipsative	Tech, Arts, Social, Fin, Trades
2	Interest Matrix	Which of these work environments sounds the most appealing?	A) Tech lab B) Hospital C) Design studio D) Boardroom E) Outdoors	Ipsative	Research, Hlth, Design, Bus, Env
3	Interest Matrix	When watching a documentary, which topic grabs your attention most?	A) AI B) Ancient history C) Human body D) Bridge building E) Startups	Ipsative	Tech, Edu, Hlth, Eng, Entr

Q#	Section	Question Text	Answer Options	Type	Trait Measured
4	Interest Matrix	Which role would you prefer in a student-run company?	A) Logo design B) Budgeting C) Pitching D) Building product E) Employee happiness	Ipsative	Arts, Fin, Bus, Tech, Social
5	Interest Matrix	I am fascinated by how the human body works and heals.	1: SD to 5: SA	Likert	Healthcare / Social
6	Interest Matrix	I enjoy debating complex rules and finding logical loopholes.	1: SD to 5: SA	Likert	Law / Enterprising
7	Interest Matrix	I would love a job where I get to work with animals or nature.	1: SD to 5: SA	Likert	Environment / Realistic
8	Interest Matrix	I frequently think about ways to start my own business.	1: SD to 5: SA	Likert	Entrepreneurship
9	Interest Matrix	I enjoy expressing myself through writing, music, or visual arts.	1: SD to 5: SA	Likert	Arts / Media
10	Interest Matrix	I like taking things apart just to see how they work.	1: SD to 5: SA	Likert	Engineering / Realistic
11	Interest Matrix	I am interested in how laws are made and government functions.	1: SD to 5: SA	Likert	Government / Law
12	Interest Matrix	I love teaching others or explaining difficult concepts.	1: SD to 5: SA	Likert	Education / Social
13	Interest Matrix	I am very interested in tracking stock markets and economies.	1: SD to 5: SA	Likert	Finance / Conventional
14	Interest Matrix	I would enjoy writing code or creating software programs.	1: SD to 5: SA	Likert	Technology
15	Interest Matrix	Which type of news article are you most likely to click on?	A) Med breakthrough B) Movie review C) Climate change D) Tax laws E) New smartphone	Ipsative	Hlth, Arts, Env, Law, Tech
16	Interest Matrix	I find deep satisfaction in organizing messy data into clean spreadsheets.	1: SD to 5: SA	Likert	Conventional / Tech
17	Interest Matrix	I want a career where my main goal is fighting for social justice.	1: SD to 5: SA	Likert	Social Impact
18	Interest Matrix	I would rather work with my hands than sit at a desk.	1: SD to 5: SA	Likert	Skilled Trades / Realistic

Q#	Section	Question Text	Answer Options	Type	Trait Measured
19	Interest Matrix	I am interested in researching space, physics, or the universe.	1: SD to 5: SA	Likert	Research / Investigative
20	Interest Matrix	I enjoy analyzing human behavior and wondering why people act the way they do.	1: SD to 5: SA	Likert	Research / Social
21	Interest Matrix	If forced to choose a club to join, which would it be?	A) Robotics B) Debate C) Drama D) Charity E) Investment	Ipsative	Tech, Law, Arts, Social, Fin
22	Interest Matrix	Which problem sounds most interesting to solve?	A) Microchip B) Genetic disease C) Energy home D) Peace treaty E) Rebrand store	Ipsative	Tech, Hlth, Eng, Gov, Bus
23	Interest Matrix	I enjoy planning out travel itineraries and schedules.	1: SD to 5: SA	Likert	Conventional / Entr
24	Interest Matrix	I love being in front of a camera or a microphone.	1: SD to 5: SA	Likert	Media / Arts
25	Interest Matrix	Which magazine cover story would you read first?	A) Quantum computing B) Top CEO C) Architecture D) Endangered Species E) Fiction Authors	Ipsative	Tech, Bus, Design, Env, Arts
26	Personality Lab	When attending a party where I don't know many people, I:	A) Intro immediately B) Wait C) Deep talk with one D) Stay near food E) Leave early	Behav.	Extroversion / Introversion
27	Personality Lab	When I have a large assignment due in two weeks, I usually:	A) Start now B) Steady schedule C) Randomly D) Wait until few days E) Panic night before	Behav.	Conscientiousness / Disc
28	Personality Lab	Unpredictable changes to my daily routine make me feel anxious.	1: SD to 5: SA	Likert	Emotional Stability / Adapt
29	Personality Lab	I consider myself a highly creative person who thinks outside the box.	1: SD to 5: SA	Likert	Openness

Q#	Section	Question Text	Answer Options	Type	Trait Measured
30	Personality Lab	I prefer to work entirely alone rather than in a group project.	1: SD to 5: SA	Likert	Team Orient / Independence
31	Personality Lab	After spending a full day socializing in a large crowd, I feel:	A) Energized B) Happy but break C) Unaffected D) Tired E) Drained	Behav.	Introversion
32	Personality Lab	I always double-check my work for small errors before submitting.	1: SD to 5: SA	Likert	Conscientiousness
33	Personality Lab	I get stressed out very easily when facing multiple deadlines.	1: SD to 5: SA	Likert	Emotional Stability
34	Personality Lab	I love trying foods from different cultures I've never heard of.	1: SD to 5: SA	Likert	Openness
35	Personality Lab	In group settings, I naturally take charge and assign roles.	1: SD to 5: SA	Likert	Leadership
36	Personality Lab	I easily forgive people when they make mistakes in a team.	1: SD to 5: SA	Likert	Agreeableness
37	Personality Lab	I keep my room and workspace highly organized at all times.	1: SD to 5: SA	Likert	Conscientiousness
38	Personality Lab	When someone criticizes my work, I:	A) Thank/apply B) Listen but hurt C) Defend D) Ignore E) Upset/doubt	Behav.	Emotional Stability
39	Personality Lab	I enjoy philosophical conversations about the meaning of life.	1: SD to 5: SA	Likert	Openness
40	Personality Lab	I find it very difficult to start conversations with strangers.	1: SD to 5: SA	Likert	Extroversion / Introversion
41	Personality Lab	I never leave tasks unfinished, even if they are boring.	1: SD to 5: SA	Likert	Discipline
42	Personality Lab	I bounce back quickly from failures or rejections.	1: SD to 5: SA	Likert	Emotional Stability
43	Personality Lab	I would rather follow a proven set of rules than invent a new way.	1: SD to 5: SA	Likert	Openness
44	Personality Lab	People often describe me as the peacemaker in my friend group.	1: SD to 5: SA	Likert	Team Orientation
45	Personality Lab	I prefer a highly predictable, structured daily schedule.	1: SD to 5: SA	Likert	Conscientiousness / Adapt

Q#	Section	Question Text	Answer Options	Type	Trait Measured
46	Personality Lab	How often do you voluntarily speak up to answer in a large class?	1: Never to 5: Always	Freq.	Extroversion
47	Personality Lab	How often do you lose important items like keys or papers?	1: Never to 5: Always	Freq.	Conscientiousness
48	Personality Lab	I am constantly curious about how the world works.	1: SD to 5: SA	Likert	Openness
49	Personality Lab	I prefer an environment where I am told exactly what to do.	1: SD to 5: SA	Likert	Independence
50	Personality Lab	I often feel overwhelmed by my own emotions.	1: SD to 5: SA	Likert	Emotional Stability
51	Future Skills	A team member stopped contributing to a crucial project. You:	A) Do it yourself B) Report them C) Message privately D) Divide work E) Submit incomplete	SJT	Prob Solving / Collab
52	Future Skills	You read a viral post making a shocking health claim. You:	A) Share immediately B) Check comments C) Search verified source D) Ignore E) Ask a friend	SJT	Crit Thinking / Dig Lit
53	Future Skills	How do you view AI tools like ChatGPT?	A) Threat B) Cheating C) Basic search D) Assistant E) Core tech to master	Scenario	AI Readiness
54	Future Skills	Given a new, complex software tool with zero training. You:	A) Refuse B) Ask colleague C) Click randomly D) YouTube tutorials E) Read manual	SJT	Learning Agility / Tech Lit
55	Future Skills	Your team suggests a solution, but you spot a major flaw. You:	A) Stay quiet B) Wait to fail C) Interrupt D) Ask guiding questions E) Fix secretly	SJT	Comm / Leadership

Q#	Section	Question Text	Answer Options	Type	Trait Measured
56	Future Skills	You notice a recurring inefficiency in organizing data. You:	A) Ignore B) Complain C) Suggest to leader D) Build auto system E) Wait for senior	SJT	Innovation / Initiative
57	Future Skills	In a fast-changing environment, I adapt my work style easily.	1: SD to 5: SA	Likert	Adaptability
58	Future Skills	When faced with a highly complex problem, my instinct is to:	A) Give up B) Guess C) Break down parts D) Look for similar E) Brainstorm wild	SJT	Problem Solving
59	Future Skills	I actively seek out feedback on how to improve my digital skills.	1: SD to 5: SA	Likert	Learning Agility
60	Future Skills	Presenting complex data to non-math people. You:	A) Read data B) Tell story C) Simple visual charts D) Tell them study E) Dense report	SJT	Comm / Creativity
61	Future Skills	I am confident in my ability to spot fake news or deepfakes.	1: SD to 5: SA	Likert	Digital Literacy
62	Future Skills	A project requires you to be highly creative. You:	A) Copy B) AI generates all C) Combine concepts D) Struggle without rules E) Unique solutions	SJT	Creativity / Innovation
63	Future Skills	I believe I will need to continuously learn new skills.	1: SD to 5: SA	Likert	Learning Agility
64	Future Skills	You disagree with a team member's approach to a task. You:	A) Argue B) Secretly do yours C) Compromise best of both D) Give in E) Complain	SJT	Collaboration
65	Future Skills	I understand how to protect my digital privacy and data security.	1: SD to 5: SA	Likert	Digital Literacy

Q#	Section	Question Text	Answer Options	Type	Trait Measured
66	Future Skills	Making a tough decision with no obvious right answer, you:	A) Coin flip B) Do what parents say C) Delay D) List pros/cons E) Gut feeling	SJT	Decision Making
67	Future Skills	I enjoy figuring out how to automate boring, repetitive tasks.	1: SD to 5: SA	Likert	AI Readiness / Innov
68	Future Skills	You are tasked with launching a totally new initiative. You:	A) Freeze B) Wait for guide C) Draft plan & test D) Research for weeks E) Ask reassignment	SJT	Initiative / Crit Thinking
69	Future Skills	I am comfortable collaborating across cultures virtually.	1: SD to 5: SA	Likert	Collaboration / Comm
70	Future Skills	If my job is automated by AI, I am confident I can pivot.	1: SD to 5: SA	Likert	Adaptability / Resilience
71	Academic Alignment	Select your STRONGEST subject area:	A) Math B) Sciences C) Languages D) Social Sci E) Arts	Pref.	Subject Strengths
72	Academic Alignment	Select your WEAKEST subject area:	A) Math B) Sciences C) Languages D) Social Sci E) Arts	Pref.	Subject Weaknesses
73	Academic Alignment	I feel confident I can succeed in highly difficult university courses.	1: SD to 5: SA	Likert	Learning Confidence
74	Academic Alignment	My current grades accurately reflect my true capability.	1: SD to 5: SA	Likert	Academic Performance
75	Academic Alignment	I am highly disciplined when studying for important exams.	1: SD to 5: SA	Likert	Study Habits
76	Academic Alignment	Which type of assessment do you perform best on?	A) Multiple choice B) Essays C) Labs D) Presentations E) Portfolios	Pref.	Learning Style
77	Academic Alignment	I struggle significantly with mathematics and numbers.	1: SD to 5: SA	Likert	Subject Weaknesses

Q#	Section	Question Text	Answer Options	Type	Trait Measured
78	Academic Alignment	I am an excellent writer and communicator in academic settings.	1: SD to 5: SA	Likert	Subject Strengths
79	Academic Alignment	I am willing to study 4-8 years to achieve my dream career.	1: SD to 5: SA	Likert	Academic Reality
80	Academic Alignment	I deeply research academic topics outside of required schoolwork.	1: SD to 5: SA	Likert	Study Habits / Curiosity
81	Contextual	I feel pressure to choose a career because my family expects me to.	1: SD to 5: SA	Likert	Parent Influence
82	Contextual	If all my friends chose the same major, I would likely choose it too.	1: SD to 5: SA	Likert	Peer Influence
83	Contextual	My parents and I have very different ideas about my future.	1: SD to 5: SA	Likert	Parent Influence / Indep
84	Contextual	Have you participated in job shadowing or workplace visits?	A) Never B) Once C) Few days D) Actively seek E) Part-time work	Freq.	Exposure to Professionals
85	Contextual	I only want a career if it guarantees a very high social status.	1: SD to 5: SA	Likert	Social Pressure
86	Contextual	I am comfortable choosing a career my family does not understand.	1: SD to 5: SA	Likert	Independent Thinking
87	Contextual	I know exactly what degree is required for my top career choice.	1: SD to 5: SA	Likert	Educational Awareness
88	Contextual	I have had a detailed conversation with an adult about their career.	1: SD to 5: SA	Likert	Career Awareness
89	Contextual	I believe society looks down on skilled trades even if they pay well.	1: SD to 5: SA	Likert	Social Pressure
90	Contextual	My choices are driven by what I see influencers doing on social media.	1: SD to 5: SA	Likert	Media Awareness
91	Motivation	When adults ask me what I want to do in the future, I feel panicked.	1: SD to 5: SA	Likert	Career Confusion
92	Motivation	I have a very clear vision of where I want to be professionally in 10 years.	1: SD to 5: SA	Likert	Career Clarity
93	Motivation	I feel highly motivated to work hard today to achieve future goals.	1: SD to 5: SA	Likert	Motivation
94	Motivation	There are so many options that I feel completely paralyzed.	1: SD to 5: SA	Likert	Career Confusion

Q#	Section	Question Text	Answer Options	Type	Trait Measured
95	Motivation	I am confident that I will be successful in whatever career I choose.	1: SD to 5: SA	Likert	Decision Confidence
96	Motivation	I frequently change my mind about what career I want to pursue.	1: SD to 5: SA	Likert	Career Clarity (Inv)
97	Motivation	Entering the adult workforce makes me feel excited rather than afraid.	1: SD to 5: SA	Likert	Motivation / Conf
98	Motivation	I believe my future career will be a major source of happiness.	1: SD to 5: SA	Likert	Motivation
99	Motivation	I feel like I don't know enough about the real world to make a decision.	1: SD to 5: SA	Likert	Career Confusion
100	Motivation	I am willing to take full ownership of my career journey.	1: SD to 5: SA	Likert	Decision Confidence / Own

PART 4: Complete Scoring Engine

Score Calculation & Weighting Rules

The platform utilizes a Weighted Sum Model.

- **Standard Likert Scales (1-5):** Assigned backend weights of 1: -2, 2: -1, 3: 0, 4: +1, 5: +2.
- **Inverted Likert Scales:** Used for negative traits (e.g., Confusion, Anxiety) weighted inversely: 1: +2 to 5: -2.
- **Ipsative / SJT Formats:** Custom weighted arrays (e.g., A: +3, B: +1, C: -1) based on the mapped trait. SJTs carry a **1.5x multiplier** to reflect higher predictive validity in applied reasoning versus self-reported statements.

Normalization Rules

All raw sums are projected onto a standardized 0-100 scale using min-max normalization, allowing dynamic comparison across domains.

$$N_t = [(S_t - S_{min}) / (S_{max} - S_{min})] \times 100$$

Composite Score Rules

- **Career Interest Scores (16 Sectors):** Summation of exact sector matches + (Sector-aligned Academic Strengths $\times$ 0.5).
- **Future Skills Score:** Weighted aggregation of AI Readiness, Critical Thinking, Digital Literacy, and Learning Agility.
- **Career Confidence Score:** (Clarity + Decision Conf + Motivation - Confusion) / 3.
- **Parent Influence Score:** Aggregation of explicit pressure questions (Q81, Q83, Q85). Scores > 70 trigger intervention flags.
- **Academic Alignment Score:** Average of normalized responses to learning confidence, study habits, and self-reported performance.
- **Entrepreneurial Potential Score:** Aggregation of Q8 + Q68 + Q100 + Risk Tolerance/Adaptability indices.
- **Leadership Score:** Aggregation of Q35 + Q55 + Q64 + Q100.

Career Cluster Rules (Applicant-Proposing Matching)

The AI calculates the mathematical overlap between the Student Trait Vector and the target requirements of various Career Clusters.

$$Match \% = [1 - (\sum w_k |S_k - C_k|) / (\sum w_k \times C_{max})] \times 100$$

(S_k = Student Trait Vector, C_k = Career Target Vector, w_k = Importance Weight).

PART 5: Student Archetype Engine

The AI runs Boolean threshold checks against normalized scores to classify the student into one of 6 Archetypes.

1. The Builder

- **Qualification Logic:** (Engineering > 70 OR Tech > 70 OR Trades > 70) AND (Problem Solving > 65) AND (Openness < 80)
- **Definition:** Pragmatic, hands-on individuals driven by tangible results, optimization, and structural creation.
- **Strengths:** High execution capability, logical reasoning, low confusion, high discipline.
- **Weaknesses:** May struggle in highly ambiguous or emotionally complex environments.
- **Career Recommendations:** Software Architect, Civil Engineer, Advanced Manufacturing, Logistics Director.

2. The Creator

- **Qualification Logic:** (Arts/Media > 75 OR Design > 75) AND (Openness > 75) AND (Creativity/Innov > 75)
- **Definition:** Innovative thinkers who excel in environments with high autonomy and abstract problem-solving.
- **Strengths:** Lateral thinking, emotional resonance, adaptability, generating novel concepts.
- **Weaknesses:** Vulnerable to burnout in structured, bureaucratic, or repetitive data roles.
- **Career Recommendations:** UX/UI Designer, Creative Director, Architecture, Multimedia Producer.

3. The Strategist

- **Qualification Logic:** (Finance > 65 OR Business > 70 OR Law > 70) AND (Conscientiousness > 70) AND (Critical Thinking > 70)
- **Definition:** Systems-level thinkers excelling at managing resources, analyzing trends, and directing teams toward long-term goals.
- **Strengths:** Organizational leadership, risk calculation, high academic alignment, resilience.
- **Weaknesses:** Analysis paralysis; may struggle with rapid, unstructured innovation.
- **Career Recommendations:** Management Consultant, Corporate Lawyer, Data Scientist, Financial Analyst.

4. The Investigator

- **Qualification Logic:** (Research > 75 OR Healthcare > 70) AND (Learning Agility > 75) AND (Introversion > 60)
- **Definition:** Deeply analytical individuals driven by the pursuit of knowledge and empirical truth.
- **Strengths:** High focus, exceptional critical thinking, mastery of complex domains.
- **Weaknesses:** Isolation in team environments; high risk of academic burnout due to perfectionism.
- **Career Recommendations:** Medical Researcher, Cybersecurity Analyst, Academic Professor, Forensic Scientist.

5. The Helper

- **Qualification Logic:** (Social Impact > 75 OR Education > 70 OR Healthcare > 65) AND (Agreeableness > 75) AND (Collaboration > 70)
- **Definition:** Empathetic, socially driven individuals focused on human development and community wellbeing.
- **Strengths:** High emotional intelligence, conflict resolution, exceptional team cohesion.
- **Weaknesses:** Compassion fatigue; struggling to assert boundaries or negotiate compensation.
- **Career Recommendations:** Clinical Psychologist, Educator, NGO Director, HR Manager, Pediatrician.

6. The Entrepreneur

- **Qualification Logic:** (Entrepreneurial Potential > 75) AND (Independence > 70) AND (Initiative > 75)
- **Definition:** High-initiative risk-takers who identify gaps in the market and mobilize resources.
- **Strengths:** Unmatched resilience, rapid learning agility, high autonomy.
- **Weaknesses:** Frustration with traditional schooling; clashes with authoritative structures.
- **Career Recommendations:** Startup Founder, Venture Capital, Growth Hacker, Product Management.

PART 6: Career Recommendation Engine

Matching Logic & Educational Pathways

The system uses the Match Percentage Logic defined in Part 4. The target vectors dictate non-negotiable baselines (e.g., Data Scientist requires Tech > 70, Logic/Math > 60, AI Readiness > 70). Matches above 85% are classified as Primary Alignments.

Once a career is matched, the engine outputs the localized Educational Pathway (e.g., High School Stream: PCM -> B.Tech Computer Science -> Certifications: AWS/Python).

Future Career Mapping

To ensure relevance, the AI heavily weights "Future Skills". Legacy career titles are appended with emerging counterparts. For example, if a student matches "Marketing" and has high AI Readiness (>80), the system appends "AI Prompt Strategist" or "Synthetic Data Engineer" to the output.

PART 7: Student Report Engine

Report Structure

The Dashboard prioritizes actionable insights over raw academic data:

1. **Executive Summary:** Dynamic archetype generation (Identity block).
2. **Global Radar:** Interactive 8-dimension mapping.
3. **Career Matches:** Expandable cards for top 3 careers with educational pathways.
4. **Future Skills Gauge:** Assessment of AI-readiness and digital agility.
5. **Action Plan:** 3 immediate, micro-steps.

Interpretation Rules (Sample Text Generation)

- **High Tech Interest:** "Deep fascination with digital systems... Strengths: Rapid tech adoption. Risks: Neglecting interpersonal skills. Improvement: Enroll in practical AI courses."
- **High Emotional Stability:** "Calm under pressure... Strengths: Crisis management. Risks: May appear detached. Improvement: Actively check in on peers."
- **Low Academic Alignment:** "Aspirations demand heavy academic rigor, but current habits are low. Improvement: Recalibrate goals or radically overhaul study habits immediately."

PART 8: Parent Report Engine

The Parent Report is an executive brief that prioritizes translation and support, actively removing academic jargon that causes parent-student friction.

Parent Guidance & Interpretation Logic

- **Low Conscientiousness** → **"Fluid Work Style"**: "Your child thrives with flexibility rather than rigid schedules. They may struggle with traditional homework routines, but can hyper-focus when a project truly captures their interest."
- **High Introversion** → **"Internal Processor"**: "Your child recharges in quiet, independent environments. They are not shy; they do their best thinking internally before speaking."
- **Low Academic Alignment** → **"Experiential Learner"**: "Current grades may not reflect true intelligence. They learn better by 'doing' rather than 'reading and memorizing.'"

Communication Suggestions

Do Not Ask: "What are you going to do with your life?" (Induces panic).

Instead Ask: "Which of the career clusters on your report surprised you the most?" (Sparks analytical conversation).

PART 9: Counselor Intelligence Engine

The backend scans the Trait Vector against 50 diagnostic rules, triggering Red/Yellow flags to guide the 1-on-1 session. Sessions follow the **Hierarchy of Needs**: Psychological Blockers > Academic Reality > Vocational Choices.

Core Counseling Rules & Intervention Logic

- **Rule 1 (High Parental Pressure):** IF Parent Influence > 75. *Flag:* High Parental Pressure. *Focus:* Explore if choices are self-directed.
- **Rule 2 (Critical Autonomy Risk):** IF Parent Influence > 75 AND Independent Thinking < 40. *Focus:* Student lacks agency; consider a mediated parent session.
- **Rule 5 (Pressure-Induced Paralysis):** IF Parent Influence > 80 AND Confusion > 70. *Focus:* Alleviate anxiety before discussing careers.
- **Rule 9 (Severe Confusion):** IF Confusion > 80. *Focus:* Abandon 10-year plans. Identify 2 broad areas of interest to research.
- **Rule 11 (Zero Industry Exposure):** IF Exposure < 20. *Focus:* Mandate job shadowing before finalizing university choices.
- **Rule 17 (Major Academic Incongruence):** IF Top Career = STEM AND Strength = Humanities. *Focus:* Confront math requirements; discuss pivot options (e.g., Tech PR).
- **Rule 19 (The Frustrated Founder):** IF Academics < 40 AND Entrepreneurial > 80. *Focus:* Validate drive; explain how basic academics protect future businesses.
- **Rule 21 (Quantitative Skill Gap):** IF Weakness = Math AND Finance > 70. *Focus:* Offer remedial pathways or adjacent careers.
- **Rule 25 (Digital Obsolescence Risk):** IF AI Readiness < 30. *Focus:* Assign basic AI literacy exploration.
- **Rule 29 (The "Idea Person" Trap):** IF Innovation > 80 AND Conscientiousness < 30. *Focus:* Teach execution frameworks and project management.
- **Rule 30 (The Lone Wolf):** IF Problem Solving > 80 AND Collaboration < 40. *Focus:* Warn that modern technical problems require team execution.
- **Rule 34 (Chronic Indecision):** IF Decision Making < 30. *Focus:* Teach that early career decisions are reversible.
- **Rule 36 (High Burnout Risk):** IF Emotional Stability < 20. *Focus:* Do NOT recommend high-stress environments regardless of interest.
- **Rule 49 (The Synthetic Creator):** IF AI Readiness > 80 AND Arts/Media > 80. *Focus:* Direct toward Generative AI media production and digital asset creation.

PART 10: AI Analysis Engine

AI Processing Logic

The AI is integrated directly via a backend Master System Prompt that orchestrates the data pipeline.

ROLE: MASTER E-BRAVE CAREER ANALYZER

You are the central cognitive engine for the E-Brave Career Discovery Platform.

INPUT PAYLOAD

JSON payload of normalized scores (0-100) across 8 dimensions.

EXECUTION INSTRUCTIONS

1. IDENTIFY ARCHETYPE: Run Boolean logic to classify into Builder, Creator, Strategist, Investigator, Helper, or Entrepreneur.
2. CAREER MAPPING: Cross-reference dominant Interests with Personality/Future Skills thresholds. Output Top 3 Careers with Educational Pathways and Emerging Roles.
3. STUDENT REPORT: Tone = Inspiring. Generate Summary, Core Strengths, Future Skills, 3-step Action Plan.
4. PARENT REPORT: Tone = Supportive. Translate lowest trait to "Fluid Work Style" or "Internal Processor".
5. COUNSELOR SUMMARY: Tone = Clinical. Evaluate 50 rules. Output Red Flags and 1-sentence "Counselor Focus".

PART 11: Product Design Blueprint

User Journeys & Dashboards

- **Student Journey:** Onboarding (Demographics/Academics) → Assessment Lab (100 Qs) → Processing Screen → **Student Dashboard** (Archetype, Global Radar, Top 3 Matches, Future Skills Gauge, Action Plan).
- **Parent Journey:** Secure Email Link → Authentication → **Parent Dashboard** (Translation Matrix, Support Strategies, Communication Guide).
- **Counselor Journey:** Login → **Master Roster** (Sorted by AI Red Flags) → Student Profile (Counselor Focus, Congruence Matrix, Session Notes Input).
- **Admin Journey:** Login → **Admin Console** (Roster Upload, Licensing, Aggregate Analytics).

Platform Workflows

Fluid, single-question interface design for the assessment. Continuous auto-saving state. Upon completion, backend triggers webhook to LLM passing normalized JSON, receives analysis, and simultaneously updates Student, Parent, and Counselor dashboards.

PART 12: Platform Architecture (Data Structures)

Implementation utilizes NoSQL documents (e.g., Firebase) or JSONB (e.g., Supabase).

1. Assessment Response JSON (Raw)

```
{
  "assessment_id": "assess_4459", "student_id": "stu_8f72",
  "status": "completed", "completion_timestamp": "2026-06-12T10:18Z",
  "raw_responses": [
    { "question_number": 1, "selected_option": "A", "time_spent_ms": 4200 }
  ]
}
```

2. Normalized Scores JSON (Calculated)

```
{
  "student_id": "stu_8f72",
  "interests": { "technology": 85, "engineering": 70, "healthcare": 30 },
  "personality": { "conscientiousness": 72, "introversion": 65 },
  "future_skills": { "ai_readiness": 92, "problem_solving": 90 },
  "contextual": { "academic_reality": 75, "parent_influence": 40 }
}
```

3. AI Analysis JSON (Master Output)

```
{
  "archetype": "The Builder",
  "top_career_matches": [ {"career_id": "car_ds_001", "match_percentage": 92.5} ],
  "counselor_flags": [
    { "rule_id": 30, "flag_title": "The Lone Wolf", "description": "Prob Solving > 80 AND Collab < 40" }
  ],
  "counselor_focus": "Student is technically brilliant but lacks collaborative skills.",
  "student_action_plan": [ "Complete an AI tutorial", "Join a team project" ]
}
```

PART 13: AntiGravity Implementation Package

Everything required to construct the platform within the AntiGravity ecosystem:

- **Pages:** /login, /student/onboarding, /student/assessment, /student/processing, /student/dashboard, /parent/dashboard, /counselor/roster, /counselor/student-view, /admin/dashboard.
- **Components:** Assessment Container (single-view, progress bar), Radar Chart Visualizer, Dynamic Future Skills Gauge, Counselor Flag Banner (Conditional render).
- **Workflows:**
 1. Q100 Submit → AntiGravity Webhook to Backend → Normalized DB Write.
 2. DB Write → LLM API Call (Master System Prompt) → Analysis DB Write.
 3. Analysis Write → SendGrid Email to Parent → Update Counselor Roster Status to "Review Ready".